

Qualifying design-stage manufacturability decisions with production evidence

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Abstract

Design-stage manufacturability decisions rest on simulation and machine learning, but whether a prediction can be trusted to decide is a property of production that is rarely measured. A production floor—the difference that drift and scatter produce between two batches of one design—is proposed as the unit; a rule that returns meets, fails or trial is characterised by its decisive distance, beyond which 95 % of its verdicts are correct, and its safe distance, beyond which at most 5 % are wrong. Six rules, from the nominal simulation to conformal, Gaussian-process and gradient-boosting corrections, are compared on 9,000 deep-drawn cups of 18 design–process alternatives with matched finite-element simulations, each alternative judged as a new design. The nominal simulation needs requirements 150 floors away to decide reliably. Calibrated on the other alternatives of its family, a Gaussian-process discrepancy model decides correctly beyond 7 floors and errs rarely beyond 2; a conformal envelope is safe beyond half a floor but decisive only beyond 25. Four to five produced alternatives that bracket the new design’s process settings make calibrated rules reliable. Across geometries no rule is safe closer than 20 floors and models without the simulation fail; the simulation carries a decision into a new family, while an applicability guard sends it to trial. Against general tolerances the best rule decides 89 % of 162 design questions correctly without a false accept. On a second process, powder bed fusion without simulation, the record of one produced orientation decides for the others within 2 floors.

Keywords: design for manufacturing, design verification, simulation credibility, uncertainty quantification, Gaussian-process calibration, deep drawing

1 Introduction

Whether a design alternative can be made to its requirements is decided long before it is made. In sheet-metal forming, as in most processes, the decision rests on a finite-element simulation of the process (Banabic, 2010): the design team simulates the alternative, compares the predicted characteristics with the requirements on the drawing and either releases the design, changes it or sends it to a physical trial. Simulation has moved such decisions forward in the design process (Thomke, 1998; Thomke and Fujimoto, 2000), and machine learning has made the loop faster and broader—surrogates replace simulations, data-driven models learn from earlier products, and manufacturability checks are increasingly automated (Yan et al, 2026; Liu et al, 2026)—but neither has changed the question a designer actually has to answer: *can this prediction be trusted to decide?* A simulation that is right on average can still be wrong for the alternative at hand, and a model that learned from earlier products may be asked about a design unlike any of them. Whitney argued in this journal that mechanical design cannot be verified by simulation the way integrated circuits are (Whitney, 1996), and variation in production remains the reason (Thornton et al, 2000): the credibility of a simulation-based decision is ultimately a property of production, not of the model, and production evidence is exactly what is missing at the design stage.

This paper treats the design-stage manufacturability decision itself as the object of study. A decision rule turns the available evidence—simulations of the new alternative and the production record of alternatives already made—into one of three verdicts: the alternative *meets* the requirement, *fails* it, or is *uncertain* and goes to a physical trial. Two questions follow: how close to the requirement can a rule still decide correctly, and how much, and which, production evidence does it need to get there? Answering them needs a unit in which the closeness of a requirement can be measured across characteristics with different physical units, and a population of produced alternatives large enough to score the verdicts against production.

The unit proposed here is the *production floor*: the difference between two batches of consecutive parts of one and the same design that drift and scatter alone produce at a chosen confidence. A difference between two alternatives smaller than the floor cannot be resolved in production, so no design-stage decision needs to be finer. Measured in floors, a decision rule has two characteristic distances: the *decisive distance*, beyond which at least 95% of its verdicts are correct, and the *safe distance*, beyond which at most 5% are wrong; between the two the rule abstains rather than errs. They are the decision-level counterpart of the limit of detection of a measurement method (Currie, 1968), and they let simulation-only, calibrated and machine-learning rules be compared on one scale.

The demonstration uses a dataset that makes such a comparison possible for the first time on open data: 9,000 deep-drawn and cut cups of dual-phase steel, produced as 18 design–process alternatives of 500 consecutive parts each, scanned after drawing and after cutting (Baum and Heinzelmann, 2026), together with finite-element simulations of the same process, including the exact tools and process settings of the experiments (Baum and Heinzelmann, 2025). Each produced alternative is treated in turn as a new

design: it is judged from its simulations and from the other alternatives’ production record, and the verdict is compared with what its 500 parts actually did.

The contributions are fourfold. The first is a definition of the production floor, with the effect-to-scatter ratio and minimum resolvable change that follow from it (Section 3.2), which tells a design team which differences between alternatives production can resolve at all. The second is a scoring of design-stage decision rules by their safe and decisive distances (Section 3.4), applied to six rules ranging from the nominal simulation through calibrated envelopes to Gaussian-process and gradient-boosting corrections, with and without the simulation, and to three calibration scopes: the same design family, both families, and a new geometry. The third is a calibration budget and calibration design: how many produced alternatives of a family a rule needs, and which ones, before its verdicts become reliable (Section 3.5), together with an applicability guard that sends designs outside the calibrated family to trial. The fourth is a six-step procedure (Section 3.6) and its demonstration on deep drawing and, without any simulation, on powder bed fusion, with findings that bear on how simulation and machine learning should be used at the design stage, robustness checks, and open code that reproduces every number from the public data.

The paper is organised as follows: Section 2 positions the work against manufacturability evaluation and machine learning in design for manufacturing, simulation credibility and calibration, and the measurement of production variation; Section 3 defines the production floor, the decision rules, the two distances, the calibration budget and the procedure; Section 4 describes the data, the characteristics and the computation; Section 5 reports the results; Section 6 discusses what they establish for design, where machine learning helps, and the threats to validity; and Section 7 concludes. Appendix A gives computation details and additional tables.

2 Background and related work

2.1 Manufacturability evaluation and machine learning in design for manufacturing

Design for manufacturing (DFM) has long codified what a process can make into rules, guidelines and cost models applied at the design stage (Boothroyd et al, 2010; Bralla, 1999), and quantitative manufacturability models have been used in this journal to drive design decisions (La Trobe-Bateman and Wild, 2003). Two developments have changed how such evaluations are made. Process simulation predicts the outcome of a specific alternative rather than of a class of features, and machine learning learns manufacturability from data: from historical designs and their production records, from simulated design spaces, or from both (Yan et al, 2026; Ballegeer and Benoit, 2026). Work in this journal has used data and models to choose between manufacturing processes from shape (Ben Slama et al, 2024), to quantify design robustness (Li et al, 2026), to reason about failure causes in design FMEA (Liu et al, 2026) and to show that measures left out of an alternative-selection problem change which alternative is chosen (Eaton et al, 2026). A recurring difficulty is transfer: a model learned on one process, machine or product family is applied to another (Safdar et al, 2024), and its reproducibility and reusability are open questions (Xie et al, 2025, 2026). What these

studies rarely have is a production record of the very alternatives the model is asked about, so the evaluation stops at prediction accuracy and does not reach the design decision; open datasets that would allow more are increasingly recognised as research contributions in their own right (Ahmed et al, 2025).

2.2 Simulation credibility and calibration

Verification and validation methodology asks whether a model is accurate enough for its intended use, and answers it by comparing predictions with experiments under a stated validation metric (Oberkampf and Roy, 2010; Roy and Oberkampf, 2011); standards for computational solid mechanics and for the credibility of decisions based on models formalise the same question (ASME, 2020; NASA, 2024). In design research, the role of physical and virtual testing in development (Tahera et al, 2019) and the evidence needed to demonstrate high reliability (Booth and Bayford, 2026) have been studied as design activities in their own right. Bayesian calibration adds a model of the discrepancy between simulator and reality, typically a Gaussian process over the inputs, and propagates its uncertainty to new predictions (Kennedy and O’Hagan, 2001; Higdon et al, 2008; Bayarri et al, 2007); omitting the discrepancy biases the calibrated parameters (Brynjarsdóttir and O’Hagan, 2014), and the identifiability of discrepancy and calibration parameters limits what can be learned from few experiments (Arendt et al, 2012). Conformal prediction gives distribution-free intervals with a finite-sample coverage guarantee under exchangeability (Vovk et al, 2005; Lei et al, 2018; Barber et al, 2021; Angelopoulos and Bates, 2023), and has begun to appear in engineering design and quality assurance (Zong et al, 2026; Heddoub et al, 2026). Both frameworks produce intervals; neither says how close to a requirement such an interval can be trusted to decide, nor how much production evidence it takes to get there.

In sheet-metal forming, the accuracy of finite-element predictions—springback above all—has been studied for decades (Wagoner et al, 2013; Chen et al, 2022), simulations are used to compensate springback in the tool design (Gan and Wagoner, 2004), and robust process design under material and process scatter is well established (Strano, 2006; Hou et al, 2010; Wiebenga et al, 2014), often with metamodels of the simulations (Jakumeit et al, 2005; Harsch et al, 2016). Data-driven models now predict forming outcomes from process and material data in series production (Havinga et al, 2020; Purr et al, 2015), and simulation datasets of deep drawing have been released for surrogate and transfer learning (Baum and Heinzelmann, 2025; Heinzelmann et al, 2025a), with surrogates of springback and part geometry trained on them (Heinzelmann et al, 2025b; Baum et al, 2025) and the gap between synthetic and real data identified as the obstacle to their industrial use (Baum et al, 2026b). The combination studied here—a production record of many alternatives with a matched simulation of each—has not been available before. The dataset’s authors decomposed the deviation between simulation and parts and found that the part geometry explains 77–92 % of its variance and that a gradient-boosting model with process measurements explains 81–92 % (Baum et al, 2026a); the present paper asks the complementary design question: how close to a requirement a decision taken before production, without process measurements of the new alternative, can be trusted.

Table 1 Positioning among methods that support design-stage manufacturability decisions

Approach	Evidence	Output	Scored against production	Abstains
DFM rules and cost models (Boothroyd et al, 2010)	codified process knowledge	guideline compliance, cost	no	no
Process simulation, V&V (Oberkampf and Roy, 2010)	physics model, validation tests	predicted characteristic, validation metric	on test cases	no
Bayesian calibration (Kennedy and O’Hagan, 2001)	simulations, experiments	predictive distribution	not as decisions	implicitly
Conformal prediction (Lei et al, 2018)	exchangeable calibration data	interval with coverage	coverage only	yes
ML manufacturability models (Yan et al, 2026)	historical or simulated data	class or value	on held-out data	rarely
This paper	simulations and produced alternatives	verdict with safe and decisive distance in production floors; calibration budget	yes, per verdict	yes

2.3 Production variation and measurement capability

Statistical quality control separates the variation of a process into short-term scatter and longer-term drift (Montgomery, 2019), and measurement systems analysis decides whether a gauge can resolve a tolerance (Automotive Industry Action Group, 2010). The limit of detection of an analytical method is the smallest quantity it detects with stated error rates (Currie, 1968, 1995), and conformity assessment under measurement uncertainty distinguishes three outcomes—conformance proven, non-conformance proven, and undecided—with guard bands set by the uncertainty (ISO, 2017; Joint Committee for Guides in Metrology, 2012). The production floor transfers the second and third ideas to design decisions: it is the resolution of production itself, measured between batches of one design, and the safe and decisive distances are detection limits of a decision rule expressed in that resolution.

Table 1 places the method among these approaches. It does not replace calibration or conformal prediction; it uses them as decision rules and scores them against production in a common unit, which is what a design team needs to choose between them.

3 Method

3.1 The decision problem

A design alternative a is a combination of product geometry and process settings that is produced as a series of parts. A quality characteristic c (a dimension, an angle, a form deviation) varies over the parts of a series. A requirement is an upper limit R on the characteristic, or on its absolute value for characteristics whose target is zero, that a share p of the parts must meet; with $p = 0.95$, alternative a is *adequate* for c

when the 95th percentile of its parts satisfies $q_{95}(a, c) \leq R$. At the design stage q_{95} is unknown. A decision rule r maps the available evidence to an interval $[L_r, U_r]$ for q_{95} and to a verdict

$$V_r = \begin{cases} \text{meets} & U_r \leq R, \\ \text{fails} & L_r > R, \\ \text{uncertain (trial)} & \text{otherwise.} \end{cases} \quad (1)$$

The three verdicts mirror the decision rules of conformity assessment, in which measurement uncertainty creates a zone where neither conformance nor non-conformance can be proven (ISO, 2017); here the uncertainty is that of the design-stage prediction. A rule that returns a point ($L_r = U_r$) always decides. The evidence consists of simulations of a and of the production record of other alternatives, the *calibration alternatives*.

3.2 Production floor, effect-to-scatter ratio and minimum resolvable change

Production evidence comes as series of consecutive parts. Each series is cut into batches of B consecutive parts, and each batch is summarised by a robust centre m (here the 10 % trimmed mean, which is insensitive to failed measurements and, unlike the median, to the quantisation of scanned edge positions). The production floor of characteristic c is

$$F_c = Q_\alpha \{ |m_i - m_j| : i \neq j \text{ batches of the same alternative} \}, \quad (2)$$

the α -quantile of the difference between two batches of one design, pooled over the produced alternatives (here $B = 50$ and $\alpha = 0.95$). Drift within a series enters the floor because batches far apart in time are compared; the floor obtained after the parts of each series are randomly permuted contains scatter only, and the ratio of the two measures how much drift adds.

The floor is the resolution of production. Two alternatives a and b are resolvable for c when their effect-to-scatter ratio

$$\text{ESR}_c(a, b) = |\bar{m}_a - \bar{m}_b| / F_c \quad (3)$$

is at least one, $\bar{m}$ being the centre of all parts of an alternative. For a continuous design or process parameter x with sensitivity $s_c = \partial c / \partial x$, the minimum resolvable change is $F_c / |s_c|$: a design iteration that changes x by less cannot be told apart in production. Uncertainty is estimated by resampling batches within alternatives; pairs of two copies of one batch are left out of a resampled floor because their difference is zero by construction.

3.3 Decision rules and calibration scopes

Six rules span the options a design team has (Table 2). M0 uses the simulation at nominal incoming conditions as it is. M1 adds the median offset between the produced

Table 2 The six decision rules

Rule	Prediction of q_{95}	Interval	Production evidence
M0 nominal simulation	simulation at nominal sheet thickness and friction	point	none
M1 bias-corrected	M0 plus the median offset	point	calibration q_{95}
M2 envelope + conformal margin	upper end of the simulation envelope plus the median offset	split-conformal, 90 %	calibration q_{95}
M5 Gaussian-process calibration	envelope plus a Gaussian-process offset over the design descriptors	predictive, 90 %	calibration q_{95}
M3 simulation + learned discrepancy	matched simulation plus a gradient-boosting model of the discrepancy, at part level	nested leave-one-out, 90 %	calibration parts
M4 learned, no simulation	gradient-boosting model of the characteristic, at part level	nested leave-one-out, 90 %	calibration parts

alternatives' q_{95} and their nominal simulation. M2 takes the upper end of the simulation envelope over plausible incoming conditions—all simulated sheet thicknesses and friction coefficients for the alternative—adds the median offset and a split-conformal margin (Lei et al, 2018) from the calibration alternatives' offsets at 90 % coverage. M5 models the offset between q_{95} and the envelope as a Gaussian process (Rasmussen and Williams, 2006) over the alternatives' descriptors, in the manner of a calibration discrepancy (Kennedy and O'Hagan, 2001), so that the correction may depend on the design and process settings, and uses its 90 % predictive interval. M3 and M4 are machine-learning rules at part level: a gradient-boosting model (Friedman, 2001) learns, on the parts of the calibration alternatives, either the discrepancy between the measured characteristic and the matched simulation (M3) or the characteristic itself without any simulation (M4), from the design descriptors and the measured incoming conditions; the predicted parts of the new alternative give its q_{95} , to which an offset and a conformal margin from a nested leave-one-out over the calibration alternatives are added. The difference between M3 and M4 is what the simulation contributes once production data are available.

Three calibration scopes describe what the design team has already produced: *within* the design family (the other alternatives with the same geometry), *pooled* (all other alternatives) and *transfer* (only alternatives of the other geometry, i.e. a new geometry is being judged).

3.4 Safe and decisive distances

The rules are scored by treating each produced alternative in turn as a new design: the rule is calibrated without it and judged at requirements placed at a known distance from the truth,

$$R = q_{95}(a, c) + d F_c, \quad (4)$$

so that a is adequate exactly when $d \geq 0$. Over all held-out alternatives and characteristics, let $\kappa(\delta)$ be the share of correct verdicts and $\omega(\delta)$ the share of wrong verdicts (false accepts and false rejects) at $|d| = \delta$. The *decisive distance* δ_d of a rule is the smallest δ beyond which $\kappa \geq 0.95$, and its *safe distance* δ_s the smallest δ beyond which

$\omega \leq 0.05$. Always $\delta_s \leq \delta_d$; in between the rule sends designs to trial rather than misleading the designer, and for a rule that always decides the two coincide. The decisive distance plays the role of a limit of detection (Currie, 1968): it is the smallest margin between prediction and requirement that the design stage can resolve. Its uncertainty is estimated by resampling the held-out alternatives (1000 resamples).

In decision-theoretic terms, a verdict costs c_{FA} when it accepts a design that production cannot make to its requirement, c_{FR} when it rejects an adequate one and c_T when it sends a design to trial. Beyond the decisive distance the expected cost of a verdict is at most $0.05 \max(c_{FA}, c_{FR}, c_T)$; beyond the safe distance it is at most $0.05 \max(c_{FA}, c_{FR}) + u c_T$, with u the share of designs sent to trial. The two distances therefore bound the cost of relying on a rule at a given requirement margin without fixing the cost ratios, which differ between products and companies; which of two rules to prefer then depends on whether a wrong verdict or a trial is the more expensive.

3.5 Calibration budget, calibration design and applicability guard

The calibration budget is the dependence of the two distances on the number k of produced alternatives of the family available for calibration, evaluated over all subsets of size k . Calibration design asks which k alternatives to produce first; subsets are classified by whether the process settings of the calibration alternatives bracket those of the new design (interpolation) or not (extrapolation). Because calibrations need not transfer across design families, an applicability guard returns *uncertain* when the new design lies outside the calibrated family: either its geometry is not represented among the calibration alternatives (family guard), or its simulated q_{95} lies outside the range of the calibration alternatives' simulated values (novelty guard), a check that uses design-stage information only.

3.6 Procedure

Figure 1 summarises the method as six steps. Step 1 frames the decision: the alternatives, the characteristics and the form of the requirement. Step 2 measures the floor on the produced alternatives, and step 3 states which differences between alternatives, and which changes of a design or process parameter, production can resolve. Step 4 calibrates the decision rules on the produced alternatives, step 5 qualifies them by their safe and decisive distances and their calibration budget, and step 6 decides for a new design: a verdict with the distance of the requirement from the prediction expressed in floors, or a trial when the guard or the interval says so.

4 Demonstration on deep drawing

4.1 Alternatives and production data

The production data are the Real Deep Drawing and Cutting dataset (RDDAC) of the University of Stuttgart (Baum and Heinzelmann, 2026; Baum et al, 2026a). Square blanks of DP600 dual-phase steel, 210 mm wide and about 1 mm thick, were deep

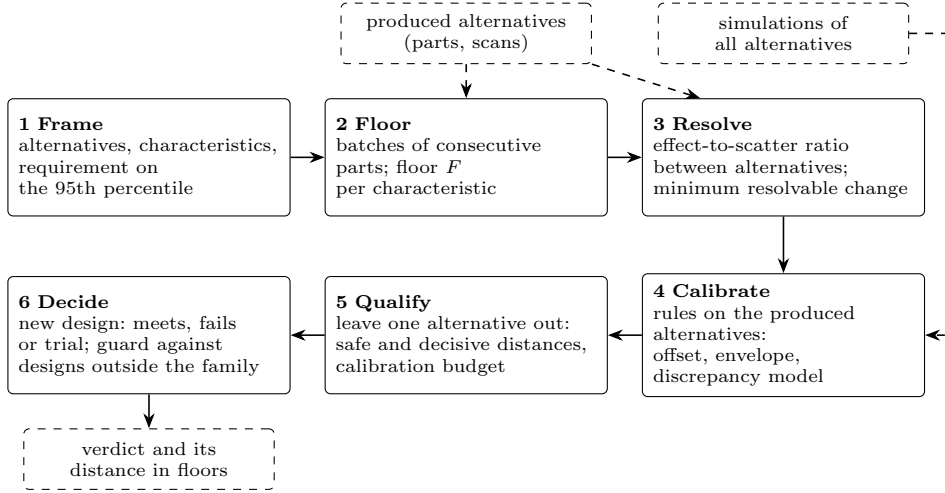


Fig. 1 The six-step procedure; dashed boxes are the evidence it uses and the verdict it returns

drawn into cups (operation OP10) and cut (OP20) on one tool set in two geometries, with concave and with convex side walls. Each geometry was run at three blank-holder forces (100, 300 and 500 kN) and three lubrication patterns (coarse, medium and fine oil application), giving 18 alternatives, each produced as one series of 500 consecutive parts. For every part the dataset holds the press force, traverses of the sheet thickness and of the oil film before forming, and 3D scans after drawing and after cutting (2000×3200 points; 0.077 mm and 0.158 mm in plane, 0.008 mm in height). Of the 9,000 parts, 10 carry no scan, leaving 8,990 for the characteristics.

4.2 Simulations

The simulations come from the Deep Drawing and Cutting Simulations dataset (DDACS) of the same group (Baum and Heinzelmann, 2025; Heinzelmann et al, 2025a): finite-element models of the same process on a quarter of the part. For the RDDAC tools, 396 simulations vary the sheet thickness (0.95–1.00 mm in six steps) and the friction coefficient (0.05–0.15 in eleven steps) at each geometry and blank-holder force; another 264 vary the geometry between two design corners per base shape at matched process settings and are used for the design-space analysis (Appendix A). A part is matched to the simulation with the nearest sheet thickness and the friction coefficient that the dataset’s linear mapping assigns to its measured oil film. The nominal simulation of an alternative is the one at 0.98 mm and friction 0.10.

4.3 Quality characteristics

Seven characteristics were extracted identically from the scans and from the simulated nodes, all referred to the plane of the cup bottom (Table 3): the draw-in of the flange at the mid-sides and at the cut corners, the waviness of the flange, the wall angle after drawing, the angle of the cut arms, the cup depth and the dome of the bottom after

Table 3 Quality characteristics, measured identically on the scans and the simulated nodes

Characteristic	Unit	Definition
Draw-in, mid-side	mm	$(2 \cdot 210 - W_x - W_y)/4$; W : flange width across the cup centre after drawing
Draw-in, corner	mm	$(2\sqrt{2} \cdot 210 - L_1 - L_2)/4$; L : distance between opposite corner tips of the flange
Flange waviness	mm	robust standard deviation of the flange about a fitted quadratic surface
Wall angle	°	inclination of the wall from the punch axis between 10 and 20 mm depth, mean of three sides
Arm angle, cut	°	inclination of the flat segment of the cut arms, mean of four arms (requirement on its absolute value)
Cup depth	mm	median depth of the flange below the plane of the cup bottom after drawing
Bottom dome, cut	mm	sag of the bottom at 40 mm radius from a fitted paraboloid (requirement on its absolute value)

cutting. The south wall of the scans, which the scanner shadows, is excluded from the wall measure.

4.4 Computation

All constants were fixed before the analysis and are listed in Appendix A.1. The pipeline reads single files from the public archives by HTTP range requests, keeps only the extracted features, and reproduces every result in this paper from the cached feature tables in about 20 minutes on four processor cores.

5 Results

5.1 Production floor

Table 4 gives the floor of each characteristic. Two batches of 50 consecutive parts of one alternative differ by up to 0.085 mm in mid-side draw-in, 0.046 mm in corner draw-in, 0.081° in wall angle, 0.11° in arm angle, 0.011 mm in cup depth and 0.007 mm in bottom dome at 95 % confidence. The floors are set by drift more than by scatter: once the time order of the parts is removed, the floor shrinks by a factor of 1.4 (mid-side draw-in, wall angle) to 4.5 (corner draw-in). Over a series the punch temperature rises by a median of 4.3 K (range 1.1–6.4 K), one of the sources of this drift; its effect is quantified in Section 5.7. For the mid-side draw-in and the wall angle the floor is smaller than the standard deviation of single parts—their parts scatter widely but their batches stay close—whereas for the other five characteristics batches drift apart by more than single parts scatter.

5.2 What production resolves

Figure 2 shows the 18 alternatives against the floor and Table 5 counts the contrasts that production resolves. All 63 geometry contrasts exceed the floor, with median effect-to-scatter ratios from 2.8 (waviness) to 130 (wall angle). Blank-holder force is

Table 4 Production floor per characteristic, pooled and per geometry

Characteristic	Floor F	95 % CI	Concave	Convex	Part SD	Drift factor
Draw-in, mid-side (mm)	0.085	0.076–0.095	0.079	0.094	0.13	1.4
Draw-in, corner (mm)	0.046	0.035–0.052	0.053	0.030	0.027	4.5
Flange waviness (mm)	0.0028	0.0024–0.0031	0.0037	0.0021	0.0020	2.5
Wall angle (°)	0.081	0.071–0.092	0.088	0.075	0.13	1.5
Arm angle, cut (°)	0.11	0.094–0.12	0.12	0.100	0.091	2.6
Cup depth (mm)	0.011	0.0096–0.012	0.0087	0.012	0.0069	3.9
Bottom dome, cut (mm)	0.0070	0.0061–0.0081	0.0034	0.0082	0.0036	3.9

Batches of 50 consecutive parts, 95 % quantile of the difference of batch centres; CI from 2000 resamples of the batches. Part SD: median within-alternative standard deviation of single parts. Drift factor: floor over the floor of randomly permuted series.

Table 5 Single-factor contrasts that production resolves and minimum resolvable force change

Characteristic	Geometry		Blank-holder force		Lubrication		Force MRC kN (95 % CI)
	ESR ≥ 1	Median	ESR ≥ 1	Median	ESR ≥ 1	Median	
Draw-in, mid-side	9/9	12.1	16/18	2.6	7/18	0.6	90 (80–100)
Draw-in, corner	9/9	108.8	18/18	4.3	7/18	0.7	55 (41–62)
Flange waviness	9/9	2.8	15/18	6.4	3/18	0.5	33 (29–37)
Wall angle	9/9	129.6	7/18	0.9	9/18	0.9	346 (289–405)
Arm angle, cut	9/9	47.2	17/18	3.0	9/18	1.0	101 (88–112)
Cup depth	9/9	34.6	18/18	10.2	7/18	0.5	28 (25–32)
Bottom dome, cut	9/9	9.6	15/18	2.4	11/18	1.6	623 (494–828)
All	63/63	34.6	106/126	3.6	53/126	0.7	

ESR: effect-to-scatter ratio of two alternatives that differ in one factor; counts of contrasts with ESR ≥ 1 and median ESR. MRC: floor over the sensitivity of the alternative centres to blank-holder force.

resolved in 106 of 126 contrasts; the exception is the wall angle, which force barely moves (7 of 18). The lubrication pattern is resolved in only 53 of 126 contrasts and in at most 11 of 18 for any characteristic: for most characteristics production cannot show the difference between the three oil-application patterns, so a design-stage decision between them has nothing to resolve. The minimum resolvable change of blank-holder force is 28 kN for the cup depth, 33 kN for the waviness, 55 kN for the corner draw-in, 90 kN for the mid-side draw-in and 101 kN for the arm angle; for the wall angle and the dome it exceeds the tested range of 400 kN. Variation of the incoming material within a series is too small to matter: moving a characteristic by one floor would take 19–147 μm of sheet thickness against a within-series standard deviation of 2.9 μm .

5.3 Simulation against production

The nominal simulation misses production by far more than a floor (Table 6). Its median offset reaches 106 floors for the cup depth of the concave cup and 108 floors for the corner draw-in of the convex cup; in units, the simulation overestimates the cup depth by about 0.9 mm and the corner draw-in by about 3 mm, and underestimates the wall angle by 0.6–1.3°. More consequential for decisions than the offset is its spread:

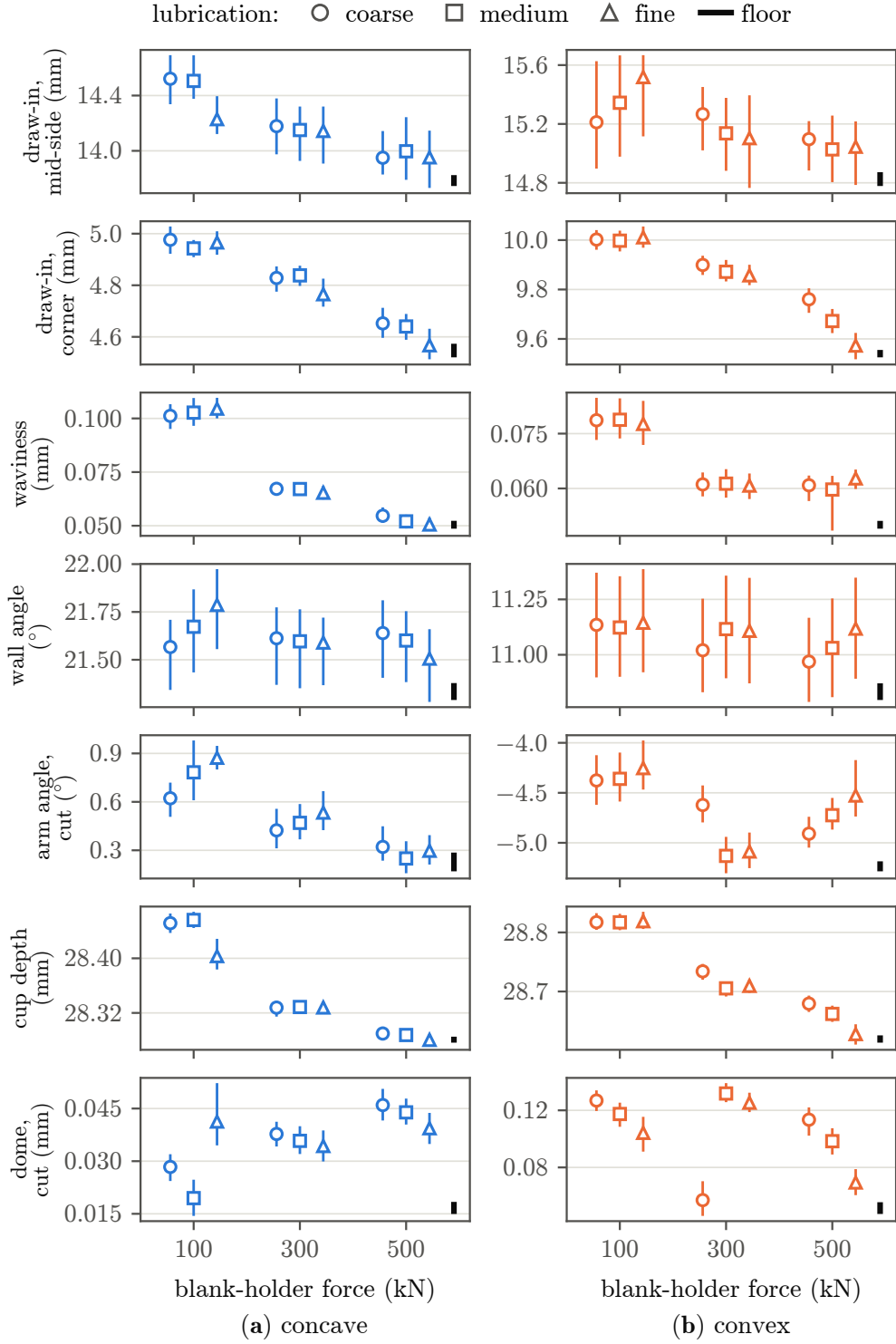


Fig. 2 Centre and 5–95 % range of each characteristic per alternative; the black bar is the production floor

Table 6 Offset between the parts’ 95th percentile and the nominal simulation, in floors

Characteristic	Concave		Convex		Matched, units	
	Offset	Spread	Offset	Spread	Concave	Convex
Draw-in, mid-side (mm)	-17	25	-16	18	-1.42	-1.76
Draw-in, corner (mm)	-59	18	-108	23	-2.96	-3.32
Flange waviness (mm)	9	14	2	11	+0.03	-0.01
Wall angle (°)	16	21	6	14	+1.25	+0.63
Arm angle, cut (°)	-30	38	1	13	-3.49	+0.04
Cup depth (mm)	-106	14	-72	30	-0.95	-0.90
Bottom dome, cut (mm)	-27	16	8	10	-0.04	+0.06

Offset: median over the nine alternatives of a geometry of $(q_{95} - \text{nominal simulation})/F$; spread: its range over the alternatives. Matched: median difference between the centre of the parts and of their matched simulations, in the characteristic’s unit.

within one geometry the offset varies by 10.5 to 38 floors between alternatives, so no constant correction can bring a decision rule below that.

5.4 Design-stage decisions

Table 7 and Fig. 3 score the six rules. The nominal simulation (M0) needs requirements 150 floors from the truth before 95 % of its verdicts are right; at 10 floors it is wrong in 31 % of the cases, mostly by rejecting adequate alternatives. Calibrated on the other eight alternatives of the family, every rule improves by an order of magnitude, in different ways. The bias correction (M1) decides everything and is right beyond 15 floors. The conformal envelope (M2) is safe beyond 0.5 floors—at 10 floors it never errs—but it sends a third of the designs to trial and decides correctly only beyond 25 floors. The Gaussian-process calibration (M5) is the most decisive rule: right beyond 7 floors (bootstrap 95 % interval 6.5–8) and safe beyond 2 floors (0.5–4.5); at 10 floors it is right in 97 % of the verdicts, never wrong, and sends 3 % to trial. Learning the characteristic from the produced alternatives without any simulation (M4) is right beyond 12 floors, and adding the simulation as the baseline of the learned discrepancy (M3) makes the rule safer but less decisive (20 floors). Within a family, the simulation therefore adds little that the production record of eight sibling alternatives does not already carry, and the best rule combines the simulation envelope with a Gaussian-process correction over the design descriptors. The intervals in Table A1 do not overlap between M5, M4, M1 and M2 within the family, so the ranking is not an artefact of the particular alternatives.

Calibrated on both geometries, M5 remains the most decisive rule (12 floors), while the extra heterogeneity widens the conformal margins of M2 and M3 (decisive at 50 and 25 floors). Calibrated only on the other geometry (Fig. 3, bottom row), no rule is safe closer than 20 floors. The rule without simulation fails outright (150 floors), because nothing in the production record of one geometry carries over to the other; with the simulation, M3 decides at 50 floors and M2 at 75, and M5, which ties with the bias correction at 40 floors, becomes over-confident: at 10 floors it is wrong in 21 % of the verdicts while abstaining in 3 %. Per characteristic (Table A2), M5 decides within

Table 7 Decisive and safe distances of the six rules in production floors

Rule	Within family		Pooled		Transfer		Within, $ d = 10$ (%)		
	δ_d	δ_s	δ_d	δ_s	δ_d	δ_s	right	wrong	trial
M0 nominal simulation	150	150	150	150	150	150	69	31	0
M1 bias-corrected	15	15	30	30	40	40	91	9	0
M2 envelope + conformal margin	25	0.5	50	0	75	20	67	0	33
M5 Gaussian-process calibration	7	2	12	1	40	30	97	0	3
M3 simulation + learned discrepancy	20	0	25	0	50	20	65	0	35
M4 learned, no simulation	12	1	25	0.5	150	150	94	0	6

Decisive (δ_d) and safe (δ_s) distances in floors over all seven characteristics and held-out alternatives; the distance grid ends at 500 floors. Bootstrap 95 % intervals over the design cases are given in Table A1; for example M5 within the family: δ_d 6.5–8, δ_s 0.5–4.5.

the family at 1.5–6.5 floors for six of the seven characteristics—roughly 0.004 mm of waviness, 0.2° of wall angle, 0.12 mm of corner draw-in, 0.38 mm of mid-side draw-in, 0.05 mm of cup depth and 0.7° of arm angle—while the dome of the bottom needs 15 floors.

5.5 Calibration budget, calibration design and guard

Figure 4 shows how the two distances depend on the number of produced alternatives of the family available for calibration. The bias correction gains little from more evidence (20 floors with one to five alternatives, 15 with six to eight), because its error is the spread of the offset between alternatives, not the uncertainty of its estimate. For the two calibrated rules, more production evidence buys safety first: with two alternatives they are wrong in 3.0–3.4 % of the verdicts at 10 floors, with four in under 1 % and with five in 0.4 %. The conformal envelope then becomes safer but not more decisive (15 floors with two alternatives, 25 from three on; safe at 0.5 floors with seven), because its margin is the largest residual of a growing calibration set. The Gaussian-process rule becomes both safer and more decisive, its decisive distance falling from 20 floors with two or three alternatives to 10 with five and 7 with seven.

Which alternatives are produced first matters as much as how many (Table 8). When the blank-holder forces of the calibration alternatives bracket that of the new design, M2 is never wrong at 10 floors from two alternatives on, and M5 in at most 1 %; when the new design lies outside them, both rules are wrong in 4.6–7.9 % and their safe distance stays at 10–12 floors. The applicability guard carries the same logic to geometry: across geometries the simulated 95th percentile of the new design lies outside the range of the calibration alternatives in 90.5 % of the cases, which go to trial; in the remaining cases M2 is never wrong at 10 floors and M5 in 2.4 %. Within a family the guard never fires, because the simulation envelope of a new alternative equals that of the calibration alternatives with the same blank-holder force.

5.6 Requirements from general tolerances

With requirements taken from general tolerances—the angular classes of ISO 2768-1 for the wall and arm angles and the flatness classes of ISO 2768-2 for the bottom

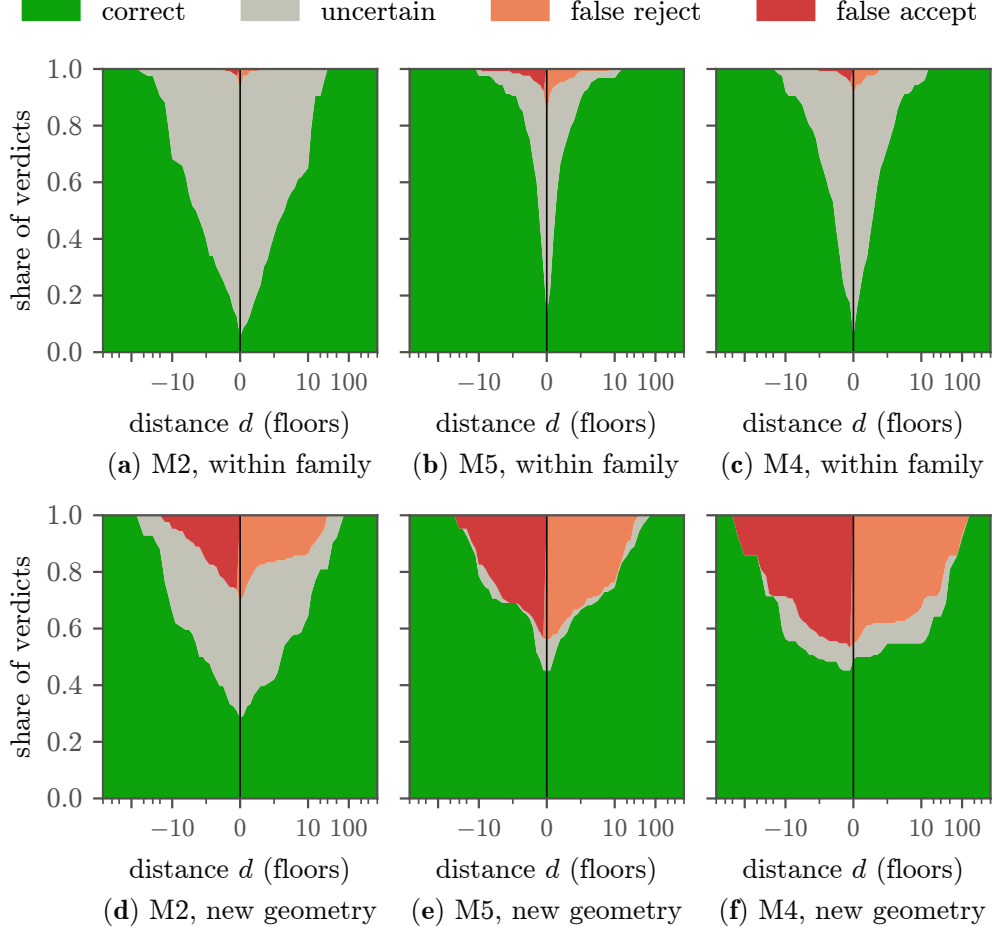


Fig. 3 Verdict shares against the requirement distance d for three rules, calibrated within the design family (top) and on the other geometry (bottom)

(ISO, 1989a,b)—the verdicts become concrete design decisions (Table 9). ISO 2768-2 addresses machined features and was withdrawn in 2021, but its classes remain a common general flatness requirement on drawings; ISO 2768-1 explicitly covers parts formed from sheet metal. The wall angle of every alternative exceeds the design angle of its tool by more than the 0.5° and 1° of classes f/m and c, and stays within the 2° of class v; the arm angle meets $\pm 1^\circ$ for the nine concave alternatives and $\pm 0.5^\circ$ for three; and 11 alternatives meet flatness class H. Within the family, M5 decides 89 % of the 162 decisions correctly, with no false accept and 1.2 % false rejects, and sends 10 % to trial; the bias correction is right in 87 % but accepts 6.8 % of the decisions in which the alternative fails; the nominal simulation is right in 66 % and accepts 19 % that fail. Across geometries, M5 is right in 65 % but falsely accepts 9.9 %, whereas M2 accepts falsely in 4.3 % and sends 39 % to trial.

Table 8 Calibration design: safe distance and wrong verdicts at 10 floors by calibration set

Rule	Calibration set	$k = 2$		$k = 3$		$k = 4$		$k = 5$	
		δ_s	wrong	δ_s	wrong	δ_s	wrong	δ_s	wrong
M1 bias-corrected	brackets	12	6	15	14	15	12	15	14
M1 bias-corrected	extrapolates	25	26	25	25	25	27	25	26
M2 envelope	brackets	3	0	1.5	0	1.5	0	1	0
M2 envelope	extrapolates	12	8	12	6	12	6	12	5
M5 GP calibration	brackets	2.5	0	3	1	2.5	0	2	0
M5 GP calibration	extrapolates	12	7	12	5	10	5	10	5

All subsets of k produced alternatives of the family; “brackets”: the blank-holder forces of the calibration alternatives enclose that of the new design. Safe distance in floors; wrong: share of wrong verdicts at $|d| = 10$ (%).

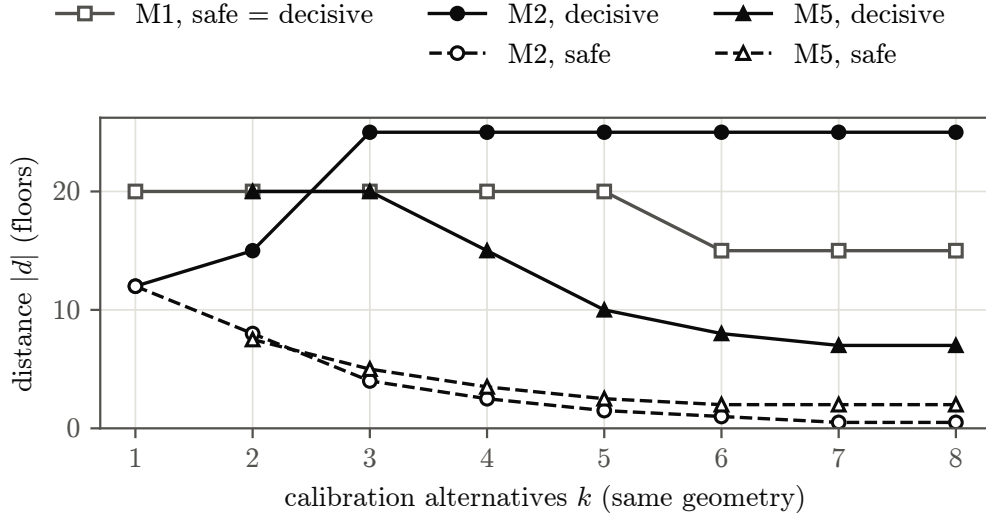


Fig. 4 Safe and decisive distances against the number of produced alternatives of the family used for calibration

5.7 Robustness

Table 10 repeats the decision analysis under variants of the data and of the constants. Removing the punch-temperature trend from every characteristic lowers the floors by 1–14 % and leaves the distances within one step of the grid (M5 within the family: 7 and 2.5 floors; pooled: 10 and 1). A conformal level of 0.80 or 0.95 instead of 0.90 leaves M2 unchanged within the family—with eight calibration alternatives its margin is the largest residual at any of these levels—and moves the decisive distance of M5 between 6 and 7.5 floors. A looser or stricter resolution target moves every distance but not the ranking: at 90 % correct M5 decides at 5 floors, M4 at 8 and M1 at 10; at 99 % at 12, 15 and 30.

The blank-holder force effects are not a by-product of the slower stroke at 100 kN: between 300 and 500 kN, at equal stroke speed, force still resolves the corner draw-in

Table 9 Verdicts against requirements from general tolerances

Rule	Within family (%)				Transfer (%)			
	right	FA	FR	trial	right	FA	FR	trial
M0 nominal simulation	66	19	15	0	66	19	15	0
M1 bias-corrected	87	7	6	0	66	14	20	0
M2 envelope + conformal margin	72	0	0	28	44	4	13	39
M5 Gaussian-process calibration	89	0	1	10	65	10	19	6
M3 simulation + learned discrepancy	63	0	1	36	51	5	13	31
M4 learned, no simulation	84	0	1	15	43	25	19	14

Nine requirements (wall angle and arm angle: ISO 2768-1 classes f/m, c, v; bottom flatness: ISO 2768-2 classes H, K, L) for 18 alternatives, 162 decisions per rule and scope. Alternatives meeting the limit: wall angle f/m 0, c 0, v 18; arm angle f/m 3, c 9, v 9; flatness H 11, K 18, L 18. FA: false accept; FR: false reject; trial: uncertain.

Table 10 Decisive/safe distances (floors) under the robustness variants

Variant	Within family			Pooled	Transfer	
	M1	M2	M5	M5	M2	M5
Reference	15	25/0.5	7/2	12/1	75/20	40/30
Temperature-adjusted characteristics	15	30/0.5	7/2.5	10/1	75/15	40/30
Conformal level 0.80	15	25/0.5	6/2.5	9/1.5	75/20	40/30
Conformal level 0.95	15	25/0.5	7.5/1.5	15/1	75/20	40/30
Resolution target 0.90	10	20/0	5/0.5	5.5/0.5	40/8.5	25/20
Resolution target 0.99	30	50/1.5	12/5.5	15/5	75/30	75/40

Temperature-adjusted: every characteristic corrected for the punch temperature with its pooled within-alternative slope, floors recomputed. Conformal level: coverage of the M2 margin and the M5 interval. Resolution target: share of correct (or at most wrong) verdicts defining the two distances. M1: decisive = safe; for the conformal variants M1 is unchanged.

and the cup depth in all six contrasts and the mid-side draw-in, the arm angle and the dome in five of six, although the sensitivity of the waviness and the arm angle flattens at the higher forces (Appendix A.2). Replacing each part’s oil film by the median film of its pattern, in the simulation match and in the model, makes M3 less decisive (40 floors instead of 20), so the within-series variation of the film carries information for the part-level model rather than noise. Finally, a conformalised Gaussian process (M6), which scales the Gaussian-process interval by the largest leave-one-out standardised residual of the calibration alternatives, is as safe as M2 (0.5 floors) but decisive only at 75 floors: with eight calibration alternatives the conformal quantile is the extreme residual, and it widens the interval of every design.

5.8 A second process: build orientation in powder bed fusion

To test whether the procedure carries over to a process without a simulation, it was applied to an open dataset on laser powder bed fusion of PA12 (Leirmo, 2021; Leirmo and Semeniuta, 2021): one test artefact printed as 135 specimens in three builds, in 37 orientations about one axis (0–180° in 5° steps), each orientation printed once in every build, and measured three times on a coordinate measuring machine. Build orientation is a genuine design-for-additive-manufacturing decision that affects dimensional and

Table 11 Build orientation in powder bed fusion of PA12: floor, resolvable class pairs and distances

Characteristic	Floor (mm)	Part SD	ESR ≥ 1	M0	M1	M2	M5
Diameter, 8 mm pin	0.13	0.13	1/15	4	1	1.5/0.5	2/0
Diameter, 24 mm pin	0.072	0.079	3/15	4	1.5	3/0.5	1/1
Diameter, 8 mm hole	0.070	0.080	2/15	4.5	1	1.5/0.5	1.5/1
Diameter, 24 mm hole	0.077	0.064	0/15	4	2	2.5/1	2/1
Hexagon across flats	0.051	0.071	6/15	6.5	3	4.5/1.5	4/1
Cylindricity, 24 mm pin	0.057	0.049	5/15	6	2	3/1	2.5/0.5
Flatness, base plate	0.072	0.073	0/15	6	1.5	2.5/0.5	2/0.5
Position, 24 mm pin	0.31	0.32	1/15	4.5	2	3.5/1	3/2.5
All			18/120	5.5	2	4/1	3.5/1

Six orientation classes of 30° , three builds; floor from the build centres of one class. ESR ≥ 1 : class pairs that production resolves. M0–M5: decisive or decisive/safe distance in floors with the class held out; no simulation is used (M0: nominal geometry).

geometric accuracy (Senthilkumaran et al, 2009; Minetola et al, 2020; Leirimo and Semeniuta, 2022). Six orientation classes of 30° serve as the alternatives and a class within one build (six or seven specimens) as a batch; eight characteristics cover the diameters of pins and holes, an across-flats distance, cylindricity, flatness and position (Table 11). Without a simulation, the design-stage evidence is the production record of the other classes: M0 predicts the nominal geometry, M1 the median 95th percentile of the other classes, M2 adds a conformal margin and M5 a Gaussian process over the orientation angle.

The floors of this process are as large as the scatter of single specimens, because the builds differ, and production resolves only 18 of the 120 pairs of orientation classes: for most characteristics a change of orientation about the tested axis is not visible against the build-to-build variation. The decision layer is correspondingly easy to satisfy. The nominal geometry is reliable beyond 5.5 floors, and the production record of the other classes decides correctly beyond 2 floors with the bias correction alone—already with a single produced class—while the interval rules are safe beyond 1 floor. The contrast with deep drawing is instructive: where the alternatives barely differ relative to the floor, the record of any one of them answers for the others, and the procedure shows it; where they differ strongly and the simulation is biased, calibration and its budget decide how close to a requirement a verdict can be trusted.

6 Discussion

6.1 What the demonstration establishes

Three results carry beyond the case. First, production has a resolution, and it is set by drift: two batches of one design differ by more than independent parts would, by a factor of 1.4 to 4.5, so the floor has to be measured on consecutive production rather than inferred from the scatter of single parts. The floor sorts design questions into those production can answer and those it cannot; here a choice between the three lubrication patterns is unresolvable for most characteristics, and changes of blank-holder force below 28–100 kN cannot be seen in five of the seven. Second, a nominal

Table 12 Guidelines for design-stage manufacturability decisions from the demonstration

Situation	Rule	What to expect and do
No produced alternative of the family	nominal simulation only for gross margins	reliable beyond about 150 floors; otherwise physical trial
Two or three produced alternatives	M2 envelope with conformal margin	choose alternatives that bracket the new design’s process settings; no wrong verdict beyond 10 floors, about a quarter sent to trial
Four or more produced alternatives	M5 Gaussian-process calibration	right beyond 7–10 floors, wrong rarely beyond 2; M2 where a false accept is costly
New geometry	M2 with the applicability guard	designs outside the simulated range of the calibration alternatives go to trial
Design or process change below the minimum resolvable change	none	not visible in production; do not iterate on it

simulation that reproduces trends is not yet a decision instrument: its offsets from production reach 108 floors and vary by 10.5 to 38 floors between alternatives of one family, so it needs requirements 150 floors away before it decides reliably. Third, with the production record of the design family the same simulation decides within a few floors, and the safe and decisive distances make the trade between deciding and abstaining explicit: the conformal envelope never misleads beyond half a floor but sends a third of the designs to trial, while the Gaussian-process calibration decides almost everything beyond 7 floors and misleads only within 2.

6.2 Implications for design practice

Table 12 turns the results into rules a design team can apply. A design-stage verdict should state its distance from the requirement in floors, so that it can be compared with the rule’s decisive and safe distances. Production evidence should be planned as part of the design of a product family: four to five produced variants make a calibrated rule reliable, and choosing them so that their process settings bracket the window in which new designs will lie removes almost all wrong verdicts—a calibration design rather than a sample size. Design iterations made on the simulation below the minimum resolvable change are invisible in production and need not be pursued. And a new geometry is a new family: no calibration from another family is safe closer than 20 floors, so the guard, not the model, should decide whether the simulation may be trusted there.

6.3 Where machine learning helps

The comparison of the six rules answers the question the call for this collection poses about learning from manufacturing data. Machine learning helps where it corrects a physical model with the production record of a design family: the Gaussian-process discrepancy is the most decisive rule within the family and pooled, and halves the decisive distance of the best rule without learning. A flexible model trained on parts

adds nothing over it: within a family a gradient-boosting model reaches 12 floors without any simulation and 20 with the simulation as its baseline, and across geometries the model without simulation fails completely because nothing in one family’s record describes the other. The simulation is therefore what carries a decision into a new design family, while learning carries the production evidence within it. Uncertainty quantification is not interchangeable either: the conformal margin is safest but indecisive with few alternatives, the Gaussian-process interval is decisive but over-confident when the alternatives are not exchangeable, and their combination did not inherit the strengths of both. A model’s interval is a statement about the model; the safe distance is a statement about production, and it is the one a designer can rely on.

6.4 Applying the procedure elsewhere

The procedure needs three things: several produced alternatives of a design family, each with a series of consecutive parts or several batches from which the floor can be measured; a design-stage predictor for every alternative, which may be a simulation, a surrogate or the production record of the other alternatives; and characteristics that can be measured identically on parts and predictions. None is specific to forming. The second demonstration applied the procedure unchanged to powder bed fusion, with builds as batches and no simulation at all, and it returned a different but equally useful answer: there, production barely separates the orientations, so the record of one of them suffices to decide for the others within 2 floors. Injection moulding with warpage simulations and machining with deflection models fit the same pattern, and the calibration budget and design apply wherever a product family is produced in variants. Where the predictor is not a physics model, rules M1 and M4 are the baselines that tell what the production record alone already achieves.

6.5 Threats to validity

The demonstration covers one material, one press and two geometries; the procedure is general, the numbers are not. Each alternative was produced once, as 500 consecutive strokes, so the floor covers drift within a series but not between days, coils or tool states and is a lower bound of the variation a requirement has to absorb in series production. The blank-holder force of 100 kN ran at a slower stroke; the force effects persist between 300 and 500 kN at equal speed, but the 100 kN alternatives combine both. Lubrication is a pattern, not a friction coefficient, and the simulations are matched through a linear mapping from the oil film that the design-space analysis calls into question; the rules M0, M1, M2 and M5 use the simulation envelope or the nominal simulation and do not depend on it. Edge positions on the scans are quantised by the pixel grid, which the trimmed-mean centres address, and absolute offsets between scans and simulations carry the scanner calibration uncertainty, which floors and effects do not. With eight to seventeen calibration alternatives the conformal margin equals the largest residual, and its guarantee assumes exchangeable alternatives, which the transfer violates by design. The requirements on the distance grid are synthetic by construction; the general-tolerance scenarios show that the conclusions hold

for requirements a drawing would carry, but not for the specification of a particular part. Finally, the design-space analysis rests on corners in which three geometric parameters change together, so single geometric effects are not identifiable from it.

7 Conclusions

Design-stage manufacturability decisions made from simulation can be scored against production, and the score is a pair of distances in a unit that production itself defines. The production floor—the difference that drift and scatter produce between two batches of one design—tells which design questions are worth asking; the decisive and safe distances tell how close to a requirement a decision rule can decide and how close it can be trusted not to mislead.

On 9,000 deep-drawn and cut cups of 18 design–process alternatives, the nominal simulation needed requirements 150 floors away before it decided reliably. Calibrated on the production record of its design family, a Gaussian-process model of the simulation’s discrepancy decided correctly beyond 7 floors and erred rarely beyond 2, and against general tolerances it decided 89 % of 162 design questions correctly without a false accept. Four to five produced alternatives that bracket the new design’s process settings made the calibrated rules reliable. Across geometries no rule was safe closer than 20 floors, a model without the simulation failed outright, and an applicability guard that uses design-stage information only sent nine in ten of these decisions to trial.

Three extensions follow: series produced on several days, coils and tool states, so that the floor covers long-term variation; more design families and processes, to turn the calibration budget into a general rule; and a cost model for trials, false accepts and false rejects, so that the choice between a safe and a decisive rule can be posed as an optimisation. The released code, which reproduces every result from the public data, is the starting point for all three.

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Declarations

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Competing interests. The author declares no competing interests.

Ethics approval and consent. Not applicable.

Data availability. All data are public. RDDAC: doi:10.18419/DARUS-5589 (version 2.0); DDACS: doi:10.18419/DARUS-4801 (version 3.0); both licensed CC BY 4.0. PA12 powder bed fusion: doi:10.18710/DHACHZ (version 1.1, CC0). The extracted per-part and per-simulation feature tables are provided with the code.

Code availability. The complete analysis pipeline is released under an open licence at <https://github.com/htcanseven/Claude> (directory `scripts/`; an archived release with a DOI will be deposited on Zenodo at acceptance) and reproduces every number in this paper from the public data.

Author contributions. H.T.C. conceived the method, designed and carried out the analysis, and wrote the paper.

Use of generative AI. A large language model (Claude, Anthropic) was used under the author’s direction as an assistant for code development, literature retrieval, data analysis and drafting and editing of the text. The author designed the study, verified all analyses, checked every reference against its published record, and takes full responsibility for the content. No AI-generated images are used; all figures are produced by the released code from the data.

Appendix A Computation details and additional tables

A.1 Computation details

Feature extraction. Scans are reduced to a grid of 2×4 -pixel medians after a luminescence filter (connected components of at least 20,040 pixels); calibration 0.0769 mm per pixel across, 0.1581 mm per line and 0.00775 mm per height unit. The cup-bottom plane is fitted robustly to the plateau at least 10 mm inside its edge, and all depths are taken from it. The flange comprises the points 20–40 mm below the bottom plane, at least 8 mm from the cup wall and 3 mm from the outer edge. Wall positions are interpolated where the binned (0.5 mm) radial profile reaches 10, 15 and 20 mm depth; the wall angle is taken between 10 and 20 mm. The arm angle is fitted to the longest run of the profile whose slope stays below 15° , shortened by 1 mm at both ends and at least 4 mm long. Flange widths are medians over bands ± 2 mm about the mid-lines; corner tips are the outline points within 0.3 mm of the extreme diagonal projection. Simulated parts are mirrored from the quarter model and processed with the same definitions on a 1 mm grid.

Floors and effects. Batches of $B = 50$ consecutive parts; batches with fewer than 80 % valid values are dropped; centres are 10 % trimmed means; $\alpha = 0.95$. Uncertainty: 2000 resamples of the batches within alternatives (seed 1), pairs of copies of one batch excluded from a resampled floor. Scatter floor: 50 random permutations of the parts within each series (seed 1).

Decision rules. Adequacy share $p = 0.95$. Requirement distances $d \in \{0, \pm 0.5, \dots, \pm 10\}$ in steps of 0.5, $\pm 12, \pm 15, \pm 20, \pm 25, \pm 30, \pm 40, \pm 50, \pm 75, \pm 100, \pm 150, \pm 200, \pm 300, \pm 500\}$ floors, per-geometry floors. Conformal margin: the $\lceil (n + 1) \cdot 0.9 \rceil$ -th smallest absolute residual of n calibration alternatives, the largest residual when this exceeds n . M5: Gaussian-process regression with a constant times anisotropic squared-exponential kernel plus white noise, normalised targets, three optimiser restarts (random state 0); descriptors are a convex-geometry indicator, the blank-holder force in units of 100 kN and the lubrication rank (coarse 0, medium 1, fine 2), with descriptors that are constant over

the calibration set dropped. M3 and M4: histogram gradient boosting (maximum depth 3, 200 iterations, learning rate 0.08, random state 0) on the geometry indicator, blank-holder force, two lubrication indicators, sheet thickness and oil film; 500 incoming conditions are drawn for the new alternative from calibration parts with the same lubrication pattern, with residuals resampled from the training fit; offset and margin from a leave-one-out over the calibration alternatives. Distance intervals: 1000 resamples of the held-out alternatives (seed 7).

Budget, guard, scenarios, robustness. The budget uses all subsets of $k = 1, \dots, 8$ calibration alternatives of the family (M5 from $k = 2$). The guards and the tolerance scenarios use the stored intervals of every rule. The temperature correction subtracts $b(T - \bar{T}_a)$ from each characteristic, with b the pooled within-alternative slope on the punch temperature T .

Software. Python 3.11 with NumPy 2.4, pandas 3.0, SciPy 1.17, scikit-learn 1.9 (Pedregosa et al, 2011) and Matplotlib 3.11.

A.2 Additional tables

Table A1 gives the bootstrap intervals of the decisive and safe distances, Table A2 the distances per characteristic within the family, Table A4 the blank-holder force contrasts with and without a change of stroke speed, Table A3 the ratio of simulated to measured sensitivity per factor, and Table A5 the in-line verifiability of the characteristics from the force record. A Gaussian-process surrogate of the simulations (Sacks et al, 1989), fitted over the process conditions at the tool geometry, reproduces them within 0.04–0.12 floors for draw-in but only within 0.6–1.9 floors for the wall angle, arm angle and depth and worse for the dome and the waviness of the convex cup, so a surrogate can stand in for the simulation only for its smooth outputs. The simulation reproduces the sensitivity of the cup depth and the dome to sheet thickness on the concave cup, overstates the effect of blank-holder force on draw-in three to seven times, and overstates the effect of friction 5 to 630 times, which points to the oil-to-friction mapping rather than to the solver. The force record explains at most 17% of the part-to-part and 11% of the batch-to-batch variation of any characteristic within an alternative, so the scan, not the press, is the evidence on which the floor rests. This does not contradict the 81–92% of the deviation between simulation and part that Baum et al (2026a) explain from process measurements: most of that deviation lies between alternatives, which the within-alternative analysis removes.

Table A1 Bootstrap 95 % intervals of the decisive and safe distances (floors)

Rule	Within family		Pooled		Transfer	
	δ_d	δ_s	δ_d	δ_s	δ_d	δ_s
M0 nominal	100–150	100–150	100–150	100–150	100–150	100–150
M1 bias-corrected	12–20	12–20	25–40	25–40	40–40	40–40
M2 envelope	25–30	0–1	40–75	0–0	75–75	12–25
M5 GP calibration	6.5–8	0.5–4.5	9–15	0.5–2.5	40–40	25–40
M3 simulation + ML	20–25	0–0.5	25–30	0–0.5	50–50	20–25
M4 ML only	10–12	0–2	15–30	0–1.5	150–200	150–200

1000 resamples of the held-out alternatives; interval ends on the distance grid.

Table A2 Decisive/safe distance per characteristic, calibrated within the family (floors)

Characteristic	M0	M1	M2	M5	M3	M4
Draw-in, mid-side	30/30	15/15	15/0.5	4.5/1	15/0.5	7.5/2
Draw-in, corner	150/150	15/15	6.5/0	2.5/0.5	15/0	10/0
Flange waviness	20/20	8/8	25/0.5	1.5/0.5	20/0.5	4/0.5
Wall angle	30/30	15/15	12/0.5	2.5/0.5	30/0	4.5/1.5
Arm angle, cut	40/40	40/40	75/0.5	6.5/2.5	20/0.5	9/1.5
Cup depth	150/150	20/20	30/0	5/3.5	20/0	12/1
Bottom dome, cut	40/40	12/12	15/1	15/9.5	30/0.5	20/4

Table A3 Simulated over measured sensitivity per factor

Characteristic	Blank-holder force		Sheet thickness		Friction	
	Conc.	Conv.	Conc.	Conv.	Conc.	Conv.
Draw-in, mid-side	4.7	6.9	-1.0	-0.66	84	142
Draw-in, corner	3.8	3.0	0.85	1.0	49	178
Flange waviness	0.64	-0.58	0.44	-4.2	-15	97
Wall angle	4.1	-2.0	2.2	2.8	5.5	-83
Arm angle, cut	7.5	0.29	11	-6.7	93	18
Cup depth	1.1	3.0	1.1	7.4	-39	69
Bottom dome, cut	0.15	0.34	1.0	-3.7	-630	21

Ratio of the slope of the characteristic on the factor in the matched simulations to the slope measured in production (blank-holder force: between cells; sheet thickness and friction equivalent of the oil film: within series). A negative ratio is an effect of the opposite sign.

Table A4 Blank-holder force contrasts with and without a change of stroke speed

Characteristic	100→300 kN (speed changes)		300→500 kN (same speed)	
	ESR ≥ 1	MRC (kN)	ESR ≥ 1	MRC (kN)
Draw-in, mid-side	5/6	75	5/6	112
Draw-in, corner	6/6	67	6/6	47
Flange waviness	6/6	21	3/6	78
Wall angle	2/6	253	2/6	544
Arm angle, cut	6/6	48	5/6	1083
Cup depth	6/6	21	6/6	43
Bottom dome, cut	5/6	543	5/6	731

Six contrasts per pair (two geometries, three lubrication patterns). MRC: pooled floor over the mean sensitivity within the pair.

Table A5 In-line verifiability of the characteristics from the press-force record

Characteristic	Part level			Batch level	
	R^2 GB	R^2 ridge	LOD/SD	R^2	Floor change (%)
Draw-in, mid-side	-0.01	0.00	3.1	-0.06	-5
Draw-in, corner	0.03	0.03	3.3	0.07	4
Flange waviness	0.02	0.01	3.2	-0.15	-6
Wall angle	-0.01	0.00	3.3	-0.08	-5
Arm angle, cut	0.00	-0.01	3.2	-0.23	-11
Cup depth	0.17	0.11	3.2	0.11	16
Bottom dome, cut	0.10	0.08	3.3	0.10	10

Out-of-fold R^2 of gradient boosting (GB) and ridge regression on force, sheet-thickness and oil-film features, folds by production batch; LOD: limit of detection of a deviation (5% false alarms, 95% detection) over the within-alternative SD. Batch level: drift between batch centres predicted with folds by alternative, and the reduction of the floor after removing it (negative: the floor grows).

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